

Daksh — Class X Study Test

Subject: Science | **Test:** Test 3 (Day 21, 30 May 2026) — Weekly
Time Allowed: 3 Hours • Maximum Marks: 80

Syllabus covered: How do Organisms Reproduce? • Heredity • Light – Reflection and Refraction

General Instructions:

- (i) This question paper contains five sections — A, B, C, D and E.
- (ii) Section A has Multiple Choice / Very Short Answer Questions of 1 mark each.
- (iii) Section B has Short Answer Questions of 2 marks each.
- (iv) Section C has Short Answer Questions of 3 marks each.
- (v) Section D has Long Answer Questions of 5 marks each.
- (vi) Section E has Case-based / Source-based questions of 4 marks each.
- (vii) All questions are compulsory. Internal choice has been provided in some questions wherever indicated.
- (viii) Use of calculators is not permitted.

SECTION A — Multiple Choice / Very Short Answer Questions (20 × 1 = 20 marks)

Answer all questions.

- Q1.** Asexual reproduction takes place through budding in: (a) Amoeba (b) Yeast (c) Plasmodium (d) Leishmania
- Q2.** Reproduction by spores is seen in: (a) Yeast (b) Hydra (c) Rhizopus (d) Planaria
- Q3.** The site of fertilisation in human female is: (a) ovary (b) uterus (c) fallopian tube (d) vagina
- Q4.** The female gamete in flowering plants is: (a) pollen grain (b) anther (c) egg cell (d) stigma
- Q5.** Which of the following is NOT a method of contraception? (a) Condom (b) IUCD (c) Tubectomy (d) Insulin
- Q6.** The unit of inheritance is: (a) chromosome (b) gene (c) nucleus (d) DNA
- Q7.** Mendel chose pea plant for his experiments because: (a) it has many traits (b) it shows clear-cut traits (c) it is easy to grow (d) all of these
- Q8.** A man with blood group A marries a woman with blood group B. The children may have which blood group? (a) only A or B (b) only AB (c) only O (d) any of A, B, AB or O
- Q9.** The sex of a human child is determined by: (a) the mother (b) the father (c) both equally (d) genes only of the mother
- Q10.** A concave mirror is used as a shaving mirror because it forms: (a) a real, inverted image (b) a virtual, erect, magnified image (c) a virtual, inverted image (d) a real, erect image
- Q11.** The focal length of a plane mirror is: (a) zero (b) infinity (c) negative (d) one metre
- Q12.** The image formed by a convex mirror is always: (a) real and inverted (b) virtual, erect, diminished (c) real and erect (d) magnified
- Q13.** The refractive index of a medium with respect to vacuum is called: (a) relative refractive index (b) absolute refractive index (c) critical angle (d) angle of incidence
- Q14.** A ray of light passes from air to water. The angle of incidence is greater than the angle of refraction because: (a) air is denser than water (b) water is optically denser than air (c) light slows down (d) both (b) and (c)
- Q15.** Power of a concave lens of focal length 20 cm is: (a) +5 D (b) –5 D (c) +0.05 D (d) –0.05 D
- Q16.** A convex lens forms a real, inverted, magnified image when the object is placed: (a) at F (b) between F and 2F (c) at 2F (d) beyond 2F
- Q17.** Define a homozygous individual.
- Q18.** Write the SI unit of power of a lens.
- Q19.** What is meant by 'pollination'?
- Q20.** State the laws of reflection of light.

SECTION B — Short Answer-I (6 × 2 = 12 marks)

Answer in 30–50 words.

- Q21.** Differentiate between sexual and asexual reproduction (any two points).
- Q22.** What is meant by vegetative propagation? Give two examples of plants reproduced by this method.
- Q23.** Explain the role of placenta in pregnancy.
- Q24.** Why is variation important for the survival of a species over a long period?
- Q25.** A real, inverted image of an object is formed by a concave mirror at a distance of 30 cm. The object is placed at 60 cm from the pole. Find the focal length of the mirror.
- Q26.** State Snell's law of refraction.

SECTION C — Short Answer-II ($7 \times 3 = 21$ marks)

Answer in 50–80 words.

- Q27.** Describe the process of fertilisation in a flowering plant. Use a labelled diagram if necessary.
- Q28.** List the parts of the human female reproductive system and state one function of each.
- Q29.** Mendel crossed a pure tall pea plant with a pure dwarf pea plant. Predict the F_1 and F_2 generations and explain Mendel's law illustrated by this experiment.
- Q30.** An object 5 cm tall is placed at a distance of 25 cm from a concave mirror of focal length 15 cm. Find the position, size and nature of the image formed.
- Q31.** What is total internal reflection? List two conditions necessary for it.
- Q32.** Define centre of curvature, principal axis and pole of a spherical mirror.
- Q33.** A convex lens has a focal length of 10 cm. An object is placed at 20 cm in front of it. Find the position, size and nature of the image.

SECTION D — Long Answer ($3 \times 5 = 15$ marks)

Answer in detail.

- Q34.** (a) Draw a labelled diagram of a flower showing the male and female reproductive parts. (3) (b) Briefly describe what happens after fertilisation in a flowering plant. (2)
- Q35.** (a) State the laws of refraction of light. (2) (b) Define refractive index. (1) (c) Light travels through glass with a speed of 2×10^8 m/s. The speed of light in vacuum is 3×10^8 m/s. Find the refractive index of glass. (2)
- Q36.** (a) State and explain Mendel's two laws of inheritance with one example each. (3) (b) Explain how the sex of a child is determined in human beings using a Punnett square. (2)

SECTION E — Case-based / Source-based ($3 \times 4 = 12$ marks)

Read each case and answer the sub-parts.

- Q37. Case Study (Reproduction).** Population growth has been a major concern for India. The government has been promoting various contraceptive methods to control birth rate.
- (i) Mention any two methods of contraception. (1)
 - (ii) How does a copper-T (IUCD) prevent pregnancy? (1)
 - (iii) Mention one disadvantage of using oral contraceptive pills. (1)
 - (iv) Why is the use of barrier methods (condoms) recommended even when other methods are available? (1)
- Q38. Case Study (Heredity).** Ravi has free-attached earlobes (recessive) while both his parents have free earlobes (dominant). Free earlobes (E) is dominant over attached earlobes (e).
- (i) What must be the genotype of Ravi? (1)
 - (ii) What are the possible genotypes of his parents? (1)
 - (iii) Explain why both parents have free earlobes but the child has attached earlobes. (2)
- Q39. Case Study (Light).** Anita uses a magnifying glass to read fine print in a newspaper. The magnifying glass is a convex lens.
- (i) Where should the newspaper be placed for the lens to act as a magnifier? (1)
 - (ii) What is the nature of the image formed in this case? (1)
 - (iii) Draw a ray diagram showing the formation of the image. (2)

— End of Question Paper —